Individual Structural Connectomes as Predictive Constraints on Macroscopic Task-State Transitions

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Abstract

This Paper 5 protocol tests whether diffusion-derived individual structural connectomes provide additional predictive constraint on macroscopic task-fMRI brain-state transitions. The primary model is G_i + S_t + X_t -> S_(t+tau), evaluated in HCP-YA 2025 Working Memory task-fMRI and diffusion data. The central comparison is P(S_(t+tau) | S_t, X_t, G_i) > P(S_(t+tau) | S_t, X_t). The analysis is frozen around a 200-subject HCP-YA 2025 cohort, with 100 development subjects and 100 locked final-holdout subjects. Current status is P5_HCP_YA_2025_DATA_NOT_AVAILABLE_LOCALLY; BALSA access probe status is LOGIN_AND_DATA_USE_TERMS_REQUIRED. The HCP structural hypothesis has not yet been tested because the required HCP-YA 2025 data packages are not available locally.

1. Introduction

Papers 3 and 4 established reproducible macroscopic stage-transition signals and tested several falsification routes. Paper 5 no longer tries to validate CWP itself. It asks a narrower biological question: whether anatomical wiring constrains which future macroscopic state follows a current task-state.

2. Methods

2.1 Dataset

The primary dataset is HCP-YA 2025. HCP-YA 2025 must not be mixed with older S1200 processing.

2.2 Cohort Lock

Target cohort: 200 eligible subjects, split into 100 development subjects and 100 locked final-holdout subjects. Eligibility requires T1w/T2w anatomy, dMRI, bvals/bvecs, WM LR/RL task-fMRI, movement regressors, and task event timing.

2.3 Structural Connectome

The frozen structural pipeline is HCP preprocessed dMRI -> MSMT-CSD -> ACT -> iFOD2 probabilistic tractography -> SIFT2 -> Schaefer-200 + subcortex parcel-by-parcel structural matrix. Primary edge weight is SIFT2-weighted connectivity. FA and streamline count are sensitivity analyses.

2.4 fMRI Brain States

Brain states are extracted from HCP Working Memory LR/RL runs using the same atlas as G_i. X_t includes 0/2-back, stimulus category, block position, run, motion, and time terms. Primary lag is 4TR_2.88s; secondary lags are 1TR_0.72s, 2TR_1.44s, 7TR_5.04s.

2.5 Models

M1: S_(t+tau) = F(S_t, X_t).

M2: S_(t+tau) = F(S_t, X_t, group_mean_G S_t).

M3: S_(t+tau) = F(S_t, X_t, G_i S_t).

2.6 Falsification

Structural nulls are swapped-subject connectomes, group-mean connectome, degree/weight-matched random graphs, edge-shuffled graphs, and distance-only graphs.

3. Current Results

3.1 E0 HCP Cohort Gate

| subject | has_t1_t2 | has_dwi | has_bvals_bvecs | has_eddy_qc | has_bedpostx | has_wm_lr | has_wm_rl | has_movement_regressors | has_wm_events | n_files | complete_for_p5 | split |

| --- | --- | --- | --- | --- | --- | --- | --- | --- | --- | --- | --- | --- |

| NO_HCP_YA_2025_FILES_FOUND | False | False | False | False | False | False | False | False | False | 0 | False | not_eligible |

3.2 E1 Structural Connectome Gate

| file | suffix | size_bytes | accepted_candidate |

| --- | --- | --- | --- |

| paper5hcp_connectomesREADME.md | .md | 396 | False |

3.3 E2-E7

E2-E7 are specified but not run because HCP-YA 2025 inputs are not local.

4. Claim Gates

Gate A: M3 > M1. Gate B: M3 > M2. Gate C: M3 > swapped/random/distance nulls. Gate D: all prior gates hold in the locked final-holdout subjects.

Current status: none of these gates has been tested.

5. Conclusion

Paper 5 has been redefined as an HCP-YA 2025 structural-connectome constraint study. The implementation package and frozen protocol are in place. The scientific hypothesis remains untested until HCP-YA 2025 task-fMRI and diffusion/connectome inputs are available locally.

Data Availability

HCP-YA 2025 data require authorized access through the HCP/ConnectomeDB/BALSA distribution path. No HCP data are redistributed here.

Code Availability

[TO BE FILLED]

Ethics Statement

This planned analysis uses public controlled-access/de-identified HCP data under the applicable HCP data-use terms.

Conflicts of Interest

[TO BE FILLED]

Claim Audit

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